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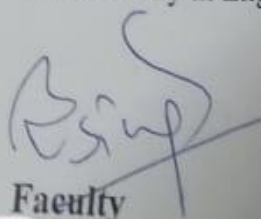
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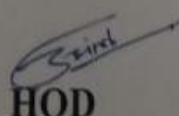
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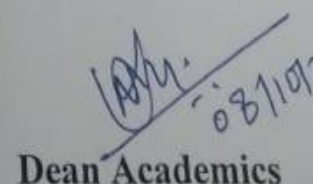
Lecture No.	Topics Covered	Page No.
1	Portland Cement; Definition	1
2	Manufacturing of cement by Rotary kiln	2-4
3	Chemistry of setting and hardening of cement, Role of Gypsum	4-6
4	Glass: Definition, Manufacturing by tank furnace, significance of annealing	6-10
5	Types and properties of soft glass, hard glass, borosilicate glass, glass wool, safety glass	10-13
6	Lubricants: Classification, Mechanism	13-16
7	Properties; Viscosity and viscosity index	16
8	Flash and fire point	16-17
9	Cloud and pour point	17-18

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1. Engineering Chemistry, by S. Rattan
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UNIT-4 New Engineering Materials

Portland Cement- Definition

Cement

Cement is construction material which possesses adhesive and cohesive properties and used for binding the building blocks, bricks, and stones.

Portland cement

Portland cement was discovered in 1824 by William Aspdin. Portland cement consist compound of lime, silica, alumina and iron. It forms a paste with water. This paste are hardens and binds aggregates (crushed rock, gravel sand etc) together to form a hard durable mass called Concrete. Portland cement is also known as “Magic powder”.

Chemical composition of Portland cement:

The composition of Portland cement should satisfy the following conditions.

Ingredient	Formula	% limit	Functions
Lime	CaO	60-67	Increases strength and setting time
Silica	SiO ₂	17-25	Gives strength and prolongs setting time
Alumina	Al ₂ O ₃	3-8	Increases strength
Iron oxide	Fe ₂ O ₃	0.5-6.0	Imparts grey color and strength
Magnesia	MgO	0.1-4.0	Excess % contributes to unsoundness.
Sulphur trioxide	SO ₃	1.2	Imparts soundness
Alkali	Na ₂ O + K ₂ O	0.5-3.0	Causes efflorescence

Manufacture of Portland cement:

Raw Material- The major raw material for manufacture of Portland cement are-

- 1) Calcareous materials (CaCO₃ containing materials such as lime stone, chalk, marble etc.
- 2) Argillaceous materials (Silicate containing materials such as clay, shall, slate, furnace slag etc.
- 3) Gypsum (CaSO₄.2 H₂O)

Manufacturing of cement by Rotary kiln

Manufacturing of Portland cement by Rotary Kiln process having following steps:

1. Mixing and crushing of raw materials (a. Dry process b. Wet process)
2. Burning
3. Grinding

Steps- Mixing and crushing of raw materials

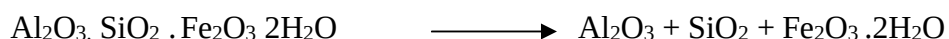
(1.a) Dry process: · In the dry process calcareous material such as lime stone (calcium carbonate) and argillaceous material such as clay are ground separately to fine powder in the absence of water and then are mixed together in the desired proportions.

(1.b) Wet process: · In the wet process, the raw materials are changed to powdered form in the presence of water. The muddy mixture formed in this process is known as slurry. The slurry is then passed into storage tanks, then slurry is then fed into rotary kiln for burning.

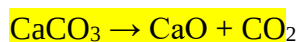
(2) Burning-

The slurry of raw material enters from the upper ends of rotary kiln while the burning fuel i.e. coal dust or oil and air are introduced from the lower ends of kiln. The slurry gradually flows in the kiln into the different zones of increasing temp.

1. **The drying zone:** The zone is present in upper part of kiln and temperature is around 400°C. Here water present in the slurry gets evaporated and clay is cleaved to Al_2O_3 , SiO_2 and Fe_2O_3 .

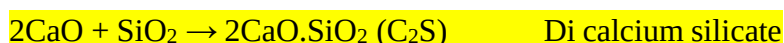


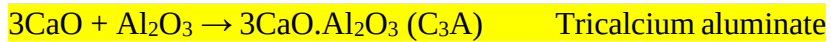
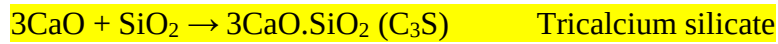
2. **Calcinations' zone or Decarbonating zone:** This is middle portion of kiln, where the temperature is 1000°C. Here the lime stone is completely decomposed into CaO which exists in the form of small lumps called nodules.



3. **Burning zone or clinkering zone:** This is bottom most and hottest zone in rotary kiln.

The temperature is around 1400°C-1500°C. In this zone the mixture melts and forms little rounded pasty mass called clinkers. The clinkers are greenish black and have rough texture. In the burning zone or clinkering zone the following reactions take place.

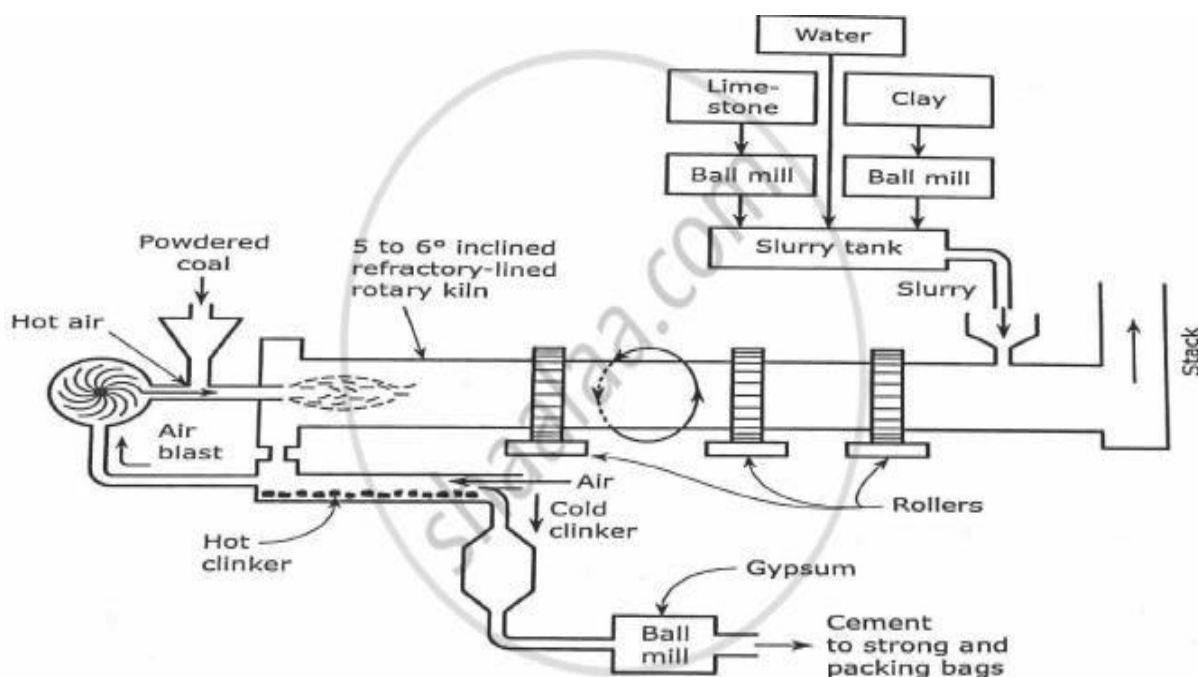




(Tetracalcium aluminoferrite)

3) **Grinding or mixing of cement clinkers with gypsum:** The clinkers are cooled and finely ground. The finely ground clinkers have low setting time and by absorption of moisture they set quickly. To reduce rate of setting it is mixed with 2-3% gypsum ($\text{CaSO}_4.2\text{H}_2\text{O}$). The fast setting Al_2O_3 in clinkers reacts with gypsum and form crystals of tricalcium sulfoaluminate which is insoluble. This prevents early setting and hardening. The mixture of clinkers and gypsum is known as Portland cement.

Compound	Chemical formula	Average percentage
Dicalcium silicate	$2\text{CaO}.\text{SiO}_2$	45%
Tricalcium silicate	$3\text{CaO}.\text{SiO}_2$	25%
Tricalcium aluminate	$2\text{CaO}.\text{Al}_2\text{O}_3$	1%
Tetracalcium aluminoferrite	$4\text{CaO} .\text{Al}_2\text{O}_3 .\text{Fe}_2\text{O}_3$	9%
Calcium Sulphate	CaSO_4	5%
Calcium oxide	CaO	62%
Magnesia	MgO	4%



MANUFACTURE OF PORTLAND CEMENT BY ROTARY KILN

Chemistry of setting and hardening of cement, Role of Gypsum

Chemistry of Setting and Hardening of Cement

Cement when mixed with water forms a plastic mass called cement paste. During hydration reaction, gel and crystalline products are formed. The inter-locking of the crystals binds the inert particles of the aggregates into a compact rock like material.

This process of solidification comprises of following steps

- (i) Initial and Final Setting
- (ii) Hardening

1) Initial and Final Setting- When cement is mixed with H_2O a plastic cement paste obtained. This paste gradually losses its plasticity and finally becomes stiff and hard. It is known as setting.

Initial setting of cement is mainly due to the hydration of tricalcium aluminates and tetra calcium alumina ferrite constituensts. Dicalcium silicate also contributes to initial setting, forming gel on hydrolysis.

Dicalcium Silicates- It hardens slowly and its effect on strength occurs in 10 days. This is due to its slower rate of reaction with water.



Tricalcium Silicates- It is highly responsible for setting and early strength as it hardens rapidly as per the reaction below.

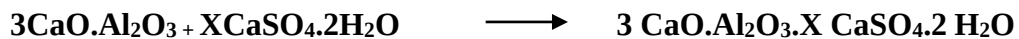


The hydal gel thus formed is responsible for the binding action between the aggregates.

Tricalcium Aluminates- This compound is firstly hydrates and gained strength within 24 hours and hydration is completed in 7 days.



The presence of gypsum in the cement helps to retards the speed of the initial set. This is due to the formation of calcium sulhoaluminate which does not show any tendency to rapid hydration.



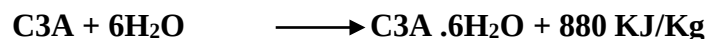
Tetracalcium alumina ferrate- It is assets in the manufacture of Portland cement by allowing low clinkering temp. It hydrates very rapidly.



Hardening- Hardening is relatively slow process after the final set, the solid begins to gain strength and it increases its strength which is called hardening. Hardening of strength depends upon the chemical combination of cement and water.

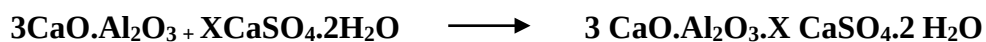
Function of gypsum in cement:-

Tri calcium aluminate (C3A) combines with water very rapidly.



After the initial setting, the paste becomes soft and the added gypsum retards the dissolution of

C3A by forming insoluble calcium sulho aluminate.



This reaction prevents the high concentration of alumina in the cement solution and hence retards the early initial setting of cement.

Glass: Definition, Manufacturing by tank furnace, significance of annealing

Glass

Glass is an amorphous, hard, brittle, transparent, super cooled, and liquid of infinite viscosity. It is obtained by fusing a mixture of sodium carbonate, calcium carbonate and sand.

Glass is a mixture of number of silicates and hence no definite chemical formula. However, it is sometimes represented as $XM_2O.y MO.SiO_2$ where M is a monovalent alkali metal like Na, K etc. M is a bivalent metal like Ca, Pb, Zn etc. And X and y are the whole number so ordinary glass may be represented as $Na_2O.CaO.6SiO_2$.

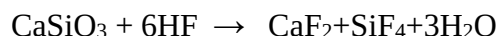
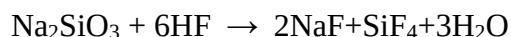
Properties of Glass

Physical properties

- 1) Glass is an amorphous, non crystalline, transparent solid and insoluble in H_2O .
- 2) It is hard and hardness depends on its composition.
- 3) Glass is a bad conductor of electricity, so it is often used as insulator.
- 4) Glass has no sharp melting point. On heating it becomes soft and changes into viscous liquid.
- 5) Glass has high viscosity.
- 6) Glass is very brittle in nature.
- 7) Glass exhibits property of vitrification (The process of conversion of molten glass to an amorphous solid instead of forming crystalline solid on slow cooling.
- 8) It can incorporate colouring material.

Chemical Properties

- 1) Glass is unreacted by all acids, except HF acid.



- 2) Ordinary glass is readily attacked by alkalies.
- 3) Glass is not attacked by air and oxidising agents.
- 4) The nature of glass is alkaline and hence it slowly reacts with water to form NaOH. This reaction is enhanced in presence of acids.

Manufacture of common glass

Raw materials

- 1) **Silica(SiO_2)**- Silica is added in the form of sand of a uniform size, but it should be free from impurities like FeO and organic matter.
- 2) **Compounds of alkali metals (Na or K)**- These are added in the form of Na_2CO_3 , Na_2SO_4 , NaNO_3 , K_2CO_3 , K_2SO_4 etc.
- 3) **Compounds of alkaline earth metals (Ca or Ba)**- The main source of calcium are lime (CaO) and limestone (CaCO_3). Barium is added in the form of BaCO_3 .
- 4) **Calcium phosphate**- Phosphate is added in the form of $\text{Ca}_3(\text{PO}_4)_2$.
- 5) **Oxides of heavy metal**- Lead is added in the form of PbO or Pb_3O_4 .
- 6) **Colouring Materials**- These metallic oxides are added to the molten glass during the manufacture of coloured glasses. Example-

Blue- Cobalt oxide

Green-Ferrous ion

Red- Cuprous oxide

Glass Manufacturing

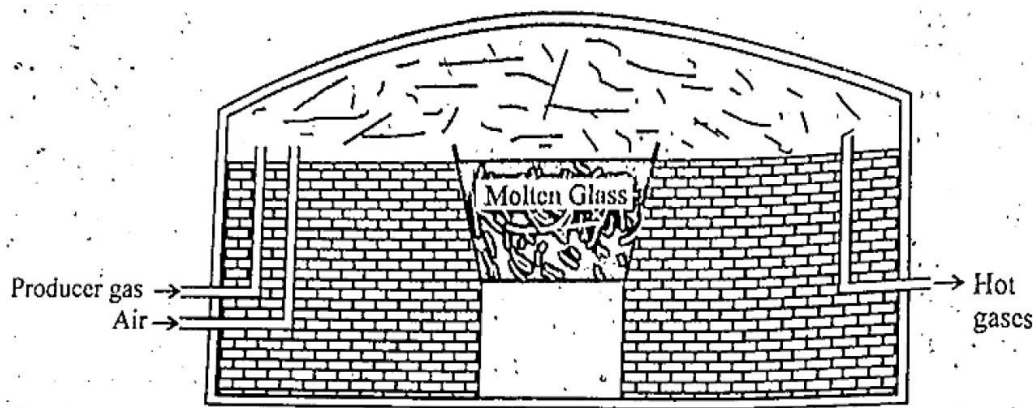
The manufacturing process of glass consists of four major operations:

(1) Melting, (2) Shaping, (3) Annealing, (4) Finishing.

1. Melting: Raw materials in proper proportions (sand, soda ash and lime stone) are mixed and finely powdered. The homogenous mixture known as Batch is fused with some broken glass, called “Cullet” in any of the two types of furnaces. The homogeneous mixture is melted either in Pot furnace or tank furnace at temp. 1800°C . Heating is done by burning producer gas and air. Cullet melts at relatively low temperature and assists in melting of rest of the charge. During the melting process following reactions occur.



The molten mass is heated till its free from air bubbles. For colored glasses desired pigment is added. The molten mass is then cooled at about 800°C at later point of time.



MANUFACTURE OF COMMON GLASS

2. Shaping: Molten glass is converted to desired shape by blowing or molding or pressing between rollers.

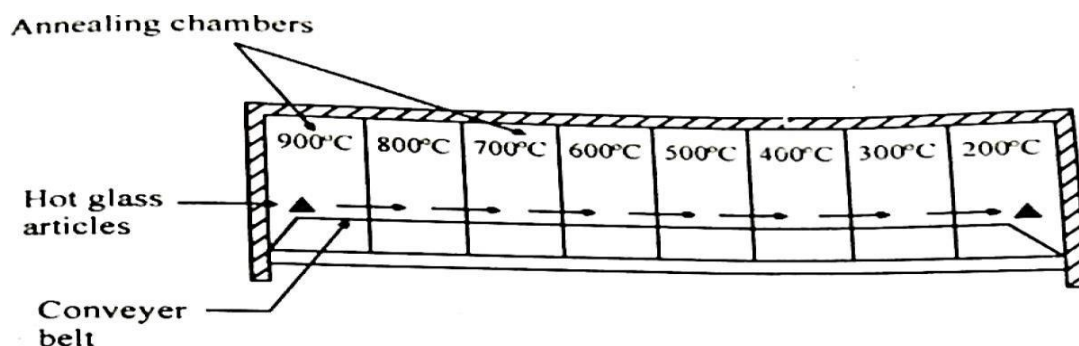
3. Annealing: The method of slow cooling of glass is called annealing. Articles are then cooled gradually at room temperature. The longer the annealing period better the quality of glass.

4. Finishing: After annealing, the articles are subjected to cleaning, polishing, cutting, sand blasting etc.,

Significance of Annealing

The method of slow cooling of glass is called annealing. If the manufactured hot glass articles are cooled suddenly, it will easily break. To avoid this glass is cooled very slowly by passing them through several hot chambers. The temperature of the hot chamber is in the decreasing order gradually. This process of slow cooling of glass is called as annealing.

Annealing is the process by which glass is made stable, for the long term, in finished form. Annealed glass will not break as long as it is not subjected to sufficient outside force.



ANNEALING PROCESS

Types and properties of soft glass, hard glass, borosilicate glass, glass wool, safety glass

Types of glasses

1. Soda-lime or soft glass
2. Potash lime or hard glass
3. Borosilicate glass or Pyrex glass or Jena glass
4. Glass wool
5. Quartz glass

Soda – lime or soft glass:

Raw Materials:

- Silica
- CaCO_3
- Soda ash

Composition : $\text{Na}_2\text{O} \cdot \text{CaO} \cdot 6\text{SiO}_2$

Properties of Soda Glass:

- Low cost
- Resistant to water
- Attacked by acids
- Melts easily
- Molded easily to any shape
- Poor thermal & chemical resistance

Uses of soda glass: Window glasses, electric bulbs, bottles, jars, table wares etc.,

2. Potash-lime or hard glass

Raw Materials

- Silica
- CaCO_3

- K_2CO_3

Composition : $K_2O \cdot CaO \cdot 6SiO_2$

Properties of Hard glass:

- High Melting point
- Not attacked by acids, alkali and other solvents
- Costlier than soda-lime glass

Uses of Hard glass:

- Combustion tubes, chemical apparatus
- Expensive to manufacture , than ordinary lime-soda glass
- Lower softening temperature than soda-lime glass
- Higher refractive index and excellent electrical properties Uses
- High quality table wares, neon sign tubings , optical lenses
- High dense glasses are used for windows to protect from X-rays and gamma rays.

Borosilicate glass or Pyrex glass or Jena glass

Raw Materials

- Silica
- Small amount of alumina
- Some oxide

Properties:

- Substitution of alkali (Na_2O) and basic alkaline earth oxides(CaO) of the soda glasses by boron and aluminium oxides results in low thermal co-efficient.
- High softening point and excellent resistivity(shock proof)
- High chemical resistance

Uses:

Industrial pipeline for corrosive liquids, gauge glasses, superior laboratory apparatus etc.

Glass Wool: Fibrous wool-like material composed of intermingled fine threads (or) filaments of glass. Alkali free. Glass filaments are obtained by forcing molten glass through small orifices which measures about 0.0005 to 0.0007 mm in diameter.

Properties

- Very good fire-proof and heat proof
- Very low electrical conductivity and thermal conductivity
- Resistance to water and most chemicals
- Tensile strength is 8X of steel

Uses

- Heat insulation purpose
- Electrical and sound insulation
- Filtration of corrosive liquids like acids
- Manufacturing fibre-glass, by blending with plastic resins.

Safety Glass:

Safety glass is glass that is specifically designed to be less likely to break, and less prone to inflicting injury when it breaks. It also includes glass that is manufactured for strength or fire resistance.

Lubricants: Classification, Mechanism

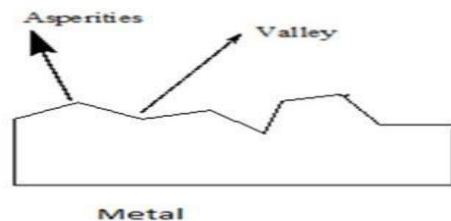
LUBRICANTS

A Lubricant is defined as a Substances which apply between two moving and sliding surface to reduce friction between them are known as Lubricants” and the process by which friction between sliding surface is reduce, known as Lubrication.

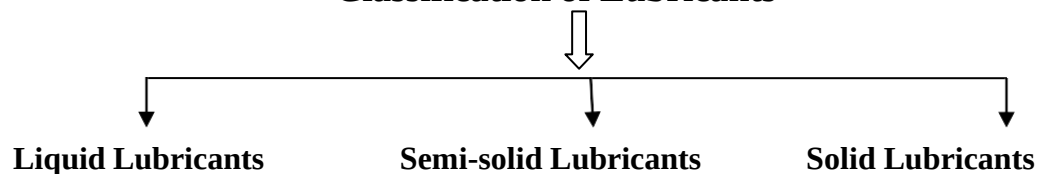
The metal surface is not smooth as its looks. In actual metals looks irregular having peaks (called asperities) and depression (called valleys). In machine, the friction is aries when metal metal surface are rubbed against each other. Due to friction large amount of energy is liberated in the form of heat which reduces the efficiency of machine.

ASPERITIES AND VALLEY

Irregularities on metal surface are called Asperities (peaks) and valley.



Classification of Lubricants



a) **Liquid Lubricants**: It includes animal oils, vegetable oils, petroleum oils, synthetic lubricants.

Animal oils - tallow oil, whale oil etc.

Vegetable oils - castor oil, palm oil etc

Petroleum oils - petroleum fractions

Synthetic lubricants -polyglycol, silicones etc.

b) Semi-solid Lubricants (Grease): Semi-solid Lubricants are formed by emulsifying oil and fat with thickening agents like soap of sodium, calcium, lithium, aluminum at higher temperature.

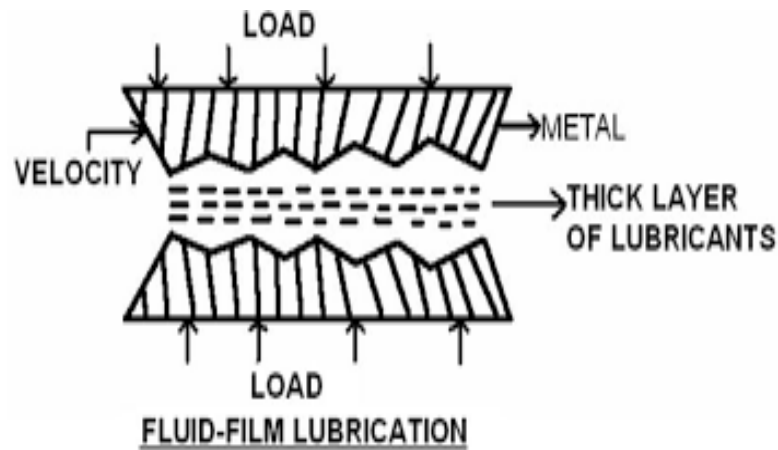
c) Solid Lubricants: Graphite, molybdenum disulphide (MoS_2), boron nitride (BN)_x are predominantly used as a solid lubricants. They are used under high temperature and high load (pressure).

i) Graphite: It is most widely used as a solid lubricant. Graphite has layer structure; layers are held together with the help of weak Vander Waals' forces which facilitate the easy sliding of one layer on the other layer. It is very soapy to touch, non-inflammable. It is used at higher temperature (around 450°C) condition. They are either used as powder form or mixed with oil or water.

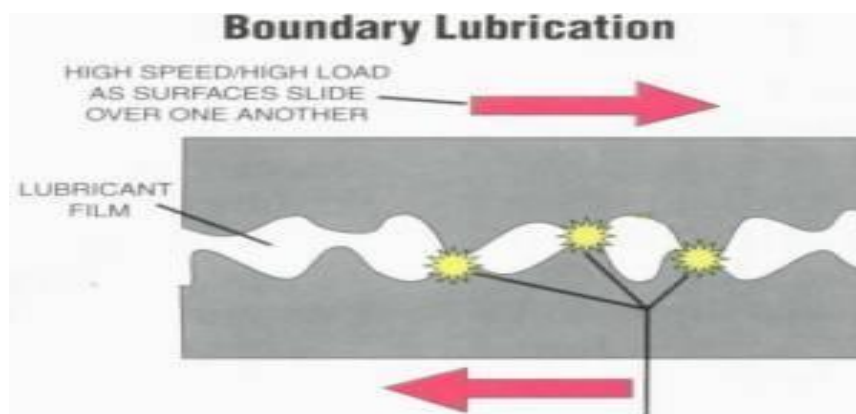
ii) Molybdenum disulphide (MoS_2): It is sandwich like structure in which hexagonal layer of molybdenum (Mo) lies between two hexagonal layers of sulfur (S) atom. Like graphite each layers are held together with weak Vander Waals' forces. It is stable up to 400°C . It adheres even more strongly to the metal or other surface.

Mechanism of Lubrication

a) Thick Film or Fluid Film or Hydrodynamic Lubrication -It is carried out with the help of liquid lubricants. In this mechanism, two moving and sliding surfaces are separated by thick film of lubricant fluid of about $1000\text{\AA}$, applied to prevent direct surface to surface contact and consequently reduce wearing and tearing of metals. Therefore it is known as thick film or fluid film lubrication or hydrodynamic (hydro meaning liquid and dynamic meaning relative motion) lubrication. In this case fluid is formed by mixing of hydrocarbon oils and anti-oxidants with long chain polymer so as to maintain viscosity. Fluid film lubrication is useful in delicate and light machines like watches, clocks, guns, scientific equipments.



- b) **Thin Film or Boundary Lubrication** - It is carried out with semi-solid (grease) and solid (graphite and molybdenum disulphide) lubricants. Boundary lubrication is a condition in which the lubricant film becomes too thin to provide total separation. In this type of lubrication a thin film of lubricant is adsorbed on the surface by weak Vander Waals forces. Thin film lubrication is operating at relatively low speed and heavy loading (pressure).



- c) **Extreme Pressure (or Temperature) Lubrication** - In this mechanism, moving or sliding surfaces are under high pressure and speed, therefore this is known as extreme pressure lubrication. In such a case high temperatures generated due to friction, under these condition liquid lubricants are fail to stick and decompose or vaporize. These problems are minimized by adding special additives to mineral oils. These additives form durable films on metal surfaces which can withstand high loads and high temperatures. Important additives are organic compound having group like chloride, sulphur, phosphorus etc. They react with metallic surface to form metallic compound (possess high melting points and serve as good lubricants under extreme temperatures and pressures) like chlorides, sulphides, phosphate as more durable film.

Viscosity and viscosity index

Viscosity - Viscosity is the property of a fluid that determines its resistance to flow. It is an indicator of flow ability of lubricating oil. The lower viscosity greater will be the flow ability. If temperature increases viscosity of the lubricating oil decreases and pressure increases viscosity of lubricating oil increases. In short we can say that good lubricating oil is that whose viscosity does not change with temperature.

Consider two layers of a liquid separated by a distance, d and moving with a relative velocity difference, v . Then, force per unit area (F) required maintaining this velocity difference is given by: $F = \eta v/d$

Where η (eta) is a constant of a liquid, called coefficient of viscosity. Viscosity is the most important single property of any lubricating oil, because it is the main determinant of the operating characteristics of the lubricant: (i) if the viscosity of the oil is too low, a liquid oil film cannot be maintained between two moving/sliding surfaces, and consequently, excessive wear will take place. On the other hand, (ii) If the viscosity is too high, excessive friction will result. The viscosity is temperature depending property.

The unit of viscosity is poise. The viscosity and velocity index of an oil is measured by using a Redwood viscometer No.1 & Redwood viscometer No.2. Both Redwood viscometer is identical and differ only in dimensions of orifice. Generally Redwood viscometer No.1 is used its orifice diameter is **1.62mm** and length of jet **10mm**

- 1. Oil Cup:** It is cylindrical brass oil cup (90 mm in height and 46.5mm in diameter) which is opened at the upper end. Oil has to be filled in it up to a certain mark. The bottom of the cup is fitted with a jet made up of 'agate' and can be opened by valve rod fitted with a ball made of brass. Thermometer is also provided to record the temperature.
- 2. Heating Bath:** The oil cup is surrounded by water bath for adjusting the temperature.
- 3. Kohlrausch's Flask:** A calibrated receiving flask (Known as Kohlrausch flask) is provided for receiving the oil from polished-agate discharged tube. Its capacity is 50 ml. up to mark in its neck.

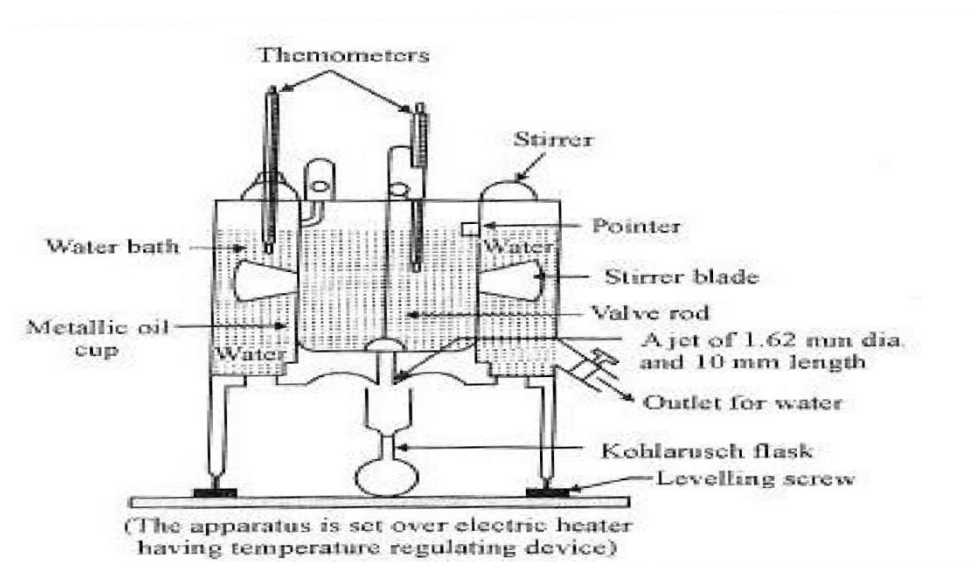


Figure 3. Redwood viscometer

Procedure:

1. Select the Redwood viscometer No.1 to determine the viscosity of given sample of lubricating oil.
2. Clean the oil cup with a sample solvent e.g. CCl_4 , ether, petroleum, spirit or benzene and dry to remove material inside.
3. Level the viscometer by leveling screws.
4. Fill heating bath with water and oil cup with oil.
5. Seal the orifice properly by means of brass ball.
6. Put the Kohlrausch's flask below the jet.
7. Maintain the desired temperature.
8. Stirrer the contents of bath cup.
9. At the equilibrium temperature, lift the ball and simultaneously start the stop watch.
10. Replace the ball and seal the cup. Refill the oil cup. Perform the experiment to get reproducible results.
11. Repeat the experiment at three different temperatures and note respective times.

d) Viscosity Index - The variation of viscosity of a liquid with temperature is called viscosity index. A relatively small change in viscosity with temperature is indicated by high viscosity index whereas, a low viscosity index shows, a relatively large change in viscosity with temperature.

Flash and fire point

Flash Point: It is the lowest temperature at which sufficient vapours are produced by the oil, which oxidize in contact of air and give a momentary flash light when a small flame is brought near it.

Fire Point: It is also the lowest temperature at which the vapours of oil begin to produce a continuous flame above the oil for at least 5 seconds when a small flame is brought near it.

Generally the fire point is 5 to 40°C higher than the flash points.

Description of Apparatus:

Pensky Marten's apparatus is a closed cup apparatus used for the determination of flash and fire points of the lubricating oil. It consists of following parts.

1. **Oil cup:** It consists of a brass cup (5 cm diameter, 5.5 cm depth) having standard size.
2. **Lid of the oil cup:** The lid has four openings, which are mean for stirrer, thermometer an air inlet and a device for introducing the standard flame.
3. **Air Bath**-Oil cup is surrounded by its flange over and air bath which is heated eletrically
4. **Electric heater:** The assembly is kept on electric heater which can be heated in a controlled manner.

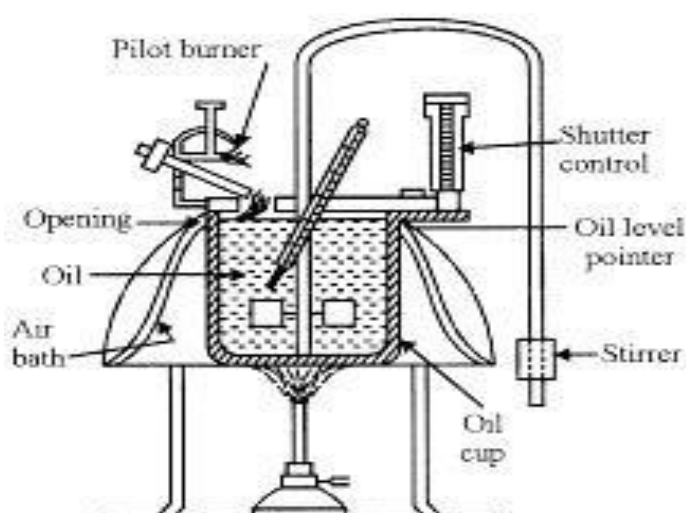


Figure1. Pensky Martens' apparatus

Procedure:

1. Clean and dry the oil cup.
2. Fill the cup with a given sample of oil.
3. Cover the lid and insert thermometer, stirrer and flame exposure device on the top.
4. The apparatus is heated so that the oil temperature increases by 5-6°C per minute while the stirrer is rotated at approx. 60 revolutions per minute.
5. When the temperature is within the range of anticipated flash point, the test flame is brought near oil vapours for about 2 sec at every degree rise of temperature.
6. Record the flash point at the temperature at the time when test flame applied causes distinct flash.
7. The oil sample is further heated at the rate of about 1°C/minute and application of test flame is done after every 1°C rise in temperature of oil. The temperature at which vapour of oil give clear and distinct flame for at least 5 seconds is recorded as its fire point.

Significance

- (i) Oil of high flash point is supposed to offer more resistance to spontaneous combustion.
- (i) The knowledge fire point of lubricating oil aids in precautionary measures against fire hazards.
- (ii) In the case of lubricating oils, it is an index of danger in using them for lubrication.

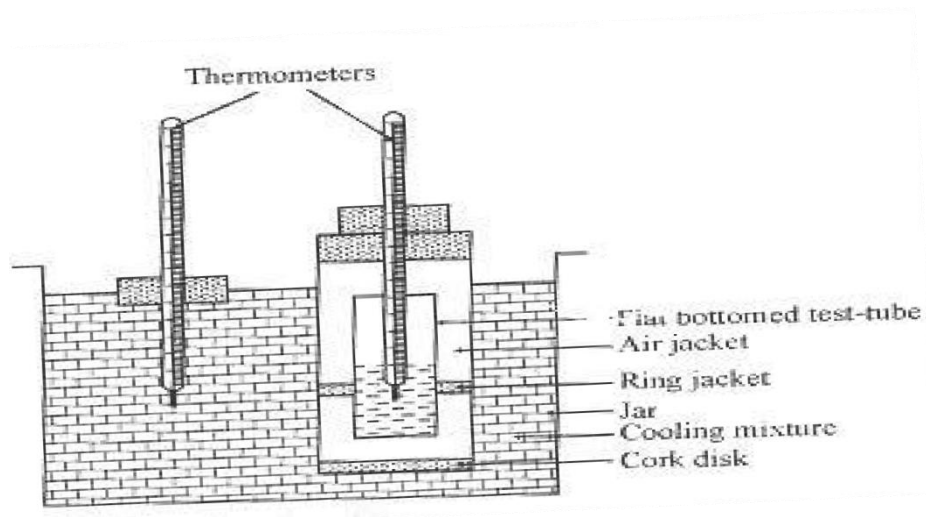
Cloud and pour point

Cloud Point: It is the lowest temperature at which lubricating oil becomes cloudy or hazy in appearance and crystallization takes place.

Pour Point: It is the lowest temperature at which the oil just ceases to flow on cooling. A high pour point may cause jamming of machines.

Cloud and pour point apparatus is used for the determination of cloud and pour points of oil. It consists of following parts.

- (i) **Test tube:** It is glass tube having flat bottom and standard dimension (3 cm diameter 12 cm height).
- (ii) **Cooling bath:** It consists of insulated jackets of glass or copper containing cooling mixture (here it is ice).
- (iii) **Thermometers:** In given sample and cooling mixture, two thermometers are fitted.



Cloud & Pour Apparatus

Procedure:

1. Thoroughly dry and clean flat bottomed glass tube.
2. Fill the oil sample and fit the cork.
3. Adjust the dipped bulb of thermometer in the oil and in the cooling bath.
4. Keep the glass tube in cooling mixture.
5. The temperature of oil falls on cooling. Take out the tube after every 1 °C fall in temperature and inspect for a moment. It may restore if the cloudiness has not appeared.
6. For cloud point, note the temperature for hazy appearance.

7. For pour point, continue the cooling process. After every 3°C fall, the tube is withdrawn and tilted to horizontal position for about 5-10 seconds. Note the temperature when oil stops flowing. It is the pour point of given oil sample.

Significance:

- (i) The cloud & pour point indicates stability of lubricants in cold conditions.
- (ii) Cloud point & pour point is useful for estimating the temperature at which filter screens in the fuel intake system of diesel engine clogged because of separation of wax.

Application of Lubricant

- i) Lubricants are primarily used to reduce the friction between two moving surface.
- ii) Rust and corrosion inhibitors
- iii) Used in the soap and paint industries.
- iv) Liquid lubricants are used in medicines
- v) Lubricants are also used as cutting fluid in cutting, grinding, drilling of metals.
- vi) Used as anti-wear, antioxidants, and antifoaming agents.